

Weekly Report

General Information

Item	Information
Group ID	Group10
Week	Week 05
Date Range	06/07/2026 – 12/07/2026

Tasks Completed This Week

Member 1

Student ID: 23127325

Full Name: Lê Gia Bảo

Completed Tasks

Designed and authored the Data Masking section for the seminar slides, including the problem statement, hybrid masking architecture (Faker.js + Gemini AI), workflow diagram, and speaker notes. Designed and developed in-class activity for data-masking, including the game rules, scoring mechanism, sample dataset, answer key strategy, and learning objectives.

Evidence

- Documents:
 - Slide_Content.md (Section # 4. Che giấu Dữ liệu - Data Masking)
 - In-class_Activi.md
-

Member 2

Student ID: 23127376

Full Name: Hồ Gia Huy

Completed Tasks

- Conducted comprehensive research on various interactive games and engaging activities suitable for the upcoming seminar session, aiming to enhance audience participation and overall engagement.
- Developed and finalized the detailed slide content for the introduction section of the presentation, ensuring a clear, captivating, and well-structured opening to effectively set the stage for the main topics.

Evidence

- Documents:
 - Slide_Content.md (Section Giới thiệu)
 - Other Artifacts:
 - Slide: database-testing-slides.html
-

Member 3

Student ID: 23127443

Full Name: Trần Phạm Trọng Nhân

Completed Tasks

- Authored slide content for the Integrity Testing (Kiểm thử Toàn vẹn) section, covering both the pain points of writing invariant tests by hand and the Agent Skill-based solution for auto-generating tests.
- Designed and documented the `test_invariants_skill.md` workflow that packages the NoSQL invariant testing procedure into a reusable AI Agent skill, including reading the real schema, placing the orphan-check in the correct referencing collection, running through `runAllInvariants.js`, and a final self-checklist.
- Supported writing the in-class activity scripts (Spot-the-PII game setup, answer-key coordinates, scoring mechanism) in coordination with the Data Masking section.

Evidence

- Documents:
 - Slide_Content.md (Session 3. Kiểm thử Toàn vẹn)
 - In-class_Activity.md
 - Other Artifacts:
 - Slide: database-testing-slides.html
 - test_invariants_skill.md
-

Member 4

Student ID: 23127503

Full Name: Trần Thanh Trí

Completed Tasks

- Write slide content for the migrate-mongo tool
- Support for writing in-class activity scripts

Evidence

- Documents:
 - Slide_Content.md (Section 2. Quản lý Phiên bản Dữ liệu)
 - In-class_Activity.md

- Other Artifacts:
 - Slide: database-testing-slides.html

AI Usage Declaration

AI Tools Used

AI Tool	Purpose	Output
Gemini (Chat)	Brainstorming interactive game ideas for the seminar, drafting slide content, and outlining the in-class activity flow	Game concepts, slide drafts, activity script
Claude (Chat)	Drafting and polishing seminar slides, expanding game ideas, and designing the Agent Skill workflow for <code>test_invariants_skill.md</code>	Slide content, game mechanics, <code>test_invariants_skill.md</code>
Cursor	Applying and validating the Agent Skill by generating invariant test files against the real MongoDB schema and running them	<code>test_invariants_skill.md</code> execution, generated test files

Description

During this week, our group utilized AI tools in the following ways to support seminar preparation and the integrity-testing workflow:

- Used Gemini Chat mainly as a brainstorming partner for the seminar: generating ideas for interactive games suitable for a classroom setting, drafting initial slide content, and shaping the storyline of the in-class activity.
- Used Claude Chat to turn those raw ideas into polished deliverables — refining the seminar slides, elaborating the game mechanics (rules, scoring, answer-key strategy), and most importantly designing the Agent Skill workflow captured in `test_invariants_skill.md`, which encodes the standard procedure for generating NoSQL invariant tests (read real schema → place orphan-check in the referencing collection → run via `runAllInvariants.js` → self-checklist).
- Used Cursor IDE to operationalize that skill: invoking the Agent to apply `test_invariants_skill.md` against the real MongoDB schemas in our codebase, inspecting the generated invariant test files, and actually running them to verify the workflow behaves correctly before committing.
- All AI-generated content — game ideas, slide drafts, skill instructions, and test code — was reviewed, refined, and verified by group members before being adopted into the repository or the seminar materials.

Note on Documentation Gaps

Due to time constraints this week, detailed logs (exact access timestamps, verbatim prompts, and screenshots/chat history) for the AI tools listed above were not fully recorded at the time of use.

We understand this does not fully satisfy the transparency requirements in Section 5 of the AI Usage Guidelines for this reporting period, and will supplement the missing details in the Week 05 report.

Tasks Planned for Next Week

- Build a lightweight web app to host the in-class activity (Spot-the-PII Game), including the CSV dataset display, cell-selection interaction, scoring logic, and answer-key coordinate validation
 - Schedule and run a full presentation rehearsal to validate timing, slide flow, and the live demo of the in-class activity web app
-

Issues

Issue	Resolution	Status
N/A	N/A	---

Notes (Optional)

None.